

Big Data Analytics of Context-Aware Itemization Effectiveness in Professional Dota 2 Matches

Professor:

Amit Patel

Team Members:

Wensheng(Wessel) Yu

Shifa Ladak

NetID:

wy889

shl8628

Course: CS-GY 6513 Big Data

Section: C

Semester: Spring 2026

Contents

1	Executive Summary	3
1.1	Project Name	3
1.2	Brief Summary	3
1.3	Objectives	3
1.4	Technologies Used	4
2	Data Sources	4
2.1	Primary Dataset	4
2.2	Supplementary Data Source	4
3	Code Execution Instructions	5
3.1	Repository and README	5
3.2	Environment Setup	5
3.3	Input Files	5
3.4	High-Level Code Logic	5
4	Methodology	6
4.1	Data Loading and Initial Processing	6
4.2	BKB Feature Construction	6
4.3	Hero Role Enrichment	6
4.4	Gold Advantage and Stomp Game Classification	6
4.5	Hero-Enemy Matchup Construction	6
4.6	Impact Metrics	7
5	Results and Analysis	7
5.1	Overall BKB Win Rate	7
5.2	BKB Impact Under Gold Advantage Context	8
5.3	Carry vs Non-Carry Heroes	8
5.4	Hero-Level BKB Impact	9
5.5	Matchup-Level BKB Analysis	10
5.6	Beyond BKB: Item Impact by Matchup	13
6	Technological Challenges	14
6.1	Parsing Item Purchase Data	14
6.2	Joining Data from Multiple Tables	14
6.3	Self-Join Expansion	14

6.4 Correlation versus Causation	15
7 Changes in Technology	15
8 Uncovered Aspects from Presentation	15
9 Lessons Learned	15
10 Future Improvements	16
11 References	16

1 Executive Summary

1.1 Project Name

Big Data Analytics of Context-Aware Itemization Effectiveness in Professional Dota 2 Matches

1.2 Brief Summary

Esports has developed into a data-intensive competitive environment where strategic decisions strongly influence match outcomes. Among esports titles, Dota 2 stands out for its high strategic complexity, including hero drafting, map control, item builds, skill progression, and player movement patterns. Among these strategic elements, itemization plays an important role because players often adjust their builds based on both their teammates and opponents.

In this project, we analyze itemization strategies in professional Dota 2 matches, with an initial focus on Black King Bar (BKB), one of the most important defensive items in the game. Instead of treating item purchase as an isolated event, we examine how item effectiveness varies across hero roles, game states, and hero-to-hero matchups.

The main idea is to move from simple win-rate comparison toward context-aware item analysis. We first compare win rates with and without BKB, then introduce additional dimensions such as gold advantage, carry-role classification, hero-specific BKB impact, and matchup-specific item impact. Finally, we extend the same framework beyond BKB to evaluate which items are associated with the highest win rate under specific hero and enemy-hero conditions.

1.3 Objectives

The project objectives are:

1. Build a scalable data processing pipeline for professional Dota 2 match data.
2. Extract player-level features such as hero, team side, match outcome, and item purchases.
3. Evaluate whether BKB is associated with higher win rate.
4. Analyze how BKB impact changes by game context, hero role, and hero matchup.
5. Extend the methodology to other items and identify matchup-specific item effectiveness.
6. Discuss limitations such as confounding variables, economic advantage, and correlation versus causation.

1.4 Technologies Used

- Python
- Pandas and NumPy
- PySpark / Spark SQL
- OpenDota API
- Seaborn and Matplotlib
- Jupyter Notebook

Although the current dataset is moderate in size, the pipeline is designed to be compatible with large-scale datasets. Key transformations such as self-joins, feature construction, groupBy aggregations, and item-level expansion can be implemented in PySpark and scaled to larger match collections.

2 Data Sources

2.1 Primary Dataset

- **Name:** Dota 2 Pro League Matches 2016–2026
- **Source:** Kaggle
- **Link:** <https://www.kaggle.com/datasets/bwandowando/dota-2-pro-league-matches>
- **Format:** CSV files
- **Main files used:** `players.csv`, `picks_bans.csv`, `radiant_gold_adv.csv`
- **Data scale:** The March 2026 player-level file contains thousands of player records. After hero-enemy self-join and item explosion, the analysis produces tens of thousands to hundreds of thousands of derived records.

2.2 Supplementary Data Source

- **Name:** OpenDota Hero Metadata API
- **Link:** <https://api.opendota.com/api/heroes>
- **Usage:** Used to map `hero_id` values to localized hero names and hero role labels such as Carry, Support, Nuker, and Disabler.

3 Code Execution Instructions

3.1 Repository and README

Code repository / README URL: <https://github.com/w3yu/Dota-2-Professional-Match-It>

3.2 Environment Setup

The project can be executed in a Python environment with Jupyter Notebook. The main required packages are:

```
pip install pandas numpy matplotlib seaborn requests pyspark
```

The Jupyter Hub for our course has these packages already installed.

3.3 Input Files

Place the dataset files in the following directory structure:

```
project_folder /
  notebook.ipynb
  202603/
    players.csv
    picks_bans.csv
    radiant_gold_adv.csv
```

3.4 High-Level Code Logic

The notebook follows this high-level pipeline:

1. Load March 2026 player-level match data from CSV.
2. Load OpenDota hero metadata and join it with player records using `hero_id`.
3. Create derived outcome variables, including player-side win labels based on `player_slot` and `radiant_win`.
4. Parse item purchase data and create item indicators such as `has_bkb`.
5. Extract final gold advantage from `radiant_goldadv.csv` and classify stomp games.
6. Use PySpark self-join on `match_id` to construct hero-enemy matchup records.
7. Compute win rates, BKB impact, and item impact by hero and matchup.

8. Generate visualizations using Seaborn and Matplotlib.

4 Methodology

4.1 Data Loading and Initial Processing

We first load the March 2026 `players.csv` dataset. Each row represents a player in a professional match and includes fields such as `match_id`, `player_slot`, `hero_id`, `radiant_win`, and item purchase information.

The win label is computed from the player's side:

- If `player_slot < 128`, the player is on Radiant.
- Otherwise, the player is on Dire.
- The player-level win value is determined by comparing the player's side with `radiant_win`.

4.2 BKB Feature Construction

The `purchase` field stores item purchase information. We parse this field and create a binary feature `has_bkb`, indicating whether the player purchased Black King Bar.

4.3 Hero Role Enrichment

To avoid manually defining hero roles, we use the OpenDota hero metadata API. A hero is classified as a carry if its role list contains the `Carry` tag. This makes the classification more consistent and data-driven.

4.4 Gold Advantage and Stomp Game Classification

The `radiant_gold_adv.csv` file contains gold advantage by match and minute. We extract the final gold advantage for each match. A stomp game is defined as a game where the winning team has more than 15,000 final gold advantage.

This definition incorporates both gold difference and match outcome to avoid misclassifying comeback games as stomp games.

4.5 Hero-Enemy Matchup Construction

To analyze opponent context, we construct hero-to-enemy records using a self-join on `match_id`. For each player, we pair that player with all five enemy heroes in the same match. This produces a matchup-level dataset where each row represents one hero facing one enemy hero.

4.6 Impact Metrics

Raw win rate can be misleading because strong heroes, favorable matchups, and economic advantage can inflate item performance. Therefore, we define item impact as:

$$\text{Item Impact} = \text{WinRate}(\text{With Item}) - \text{WinRate}(\text{Without Item}) \quad (1)$$

For BKB specifically:

$$\text{BKB Impact} = \text{WinRate}(\text{With BKB}) - \text{WinRate}(\text{Without BKB}) \quad (2)$$

This metric is not a causal estimate, but it provides a better first-order approximation than raw win rate alone.

5 Results and Analysis

5.1 Overall BKB Win Rate

We first compare player win rate between records with and without BKB.

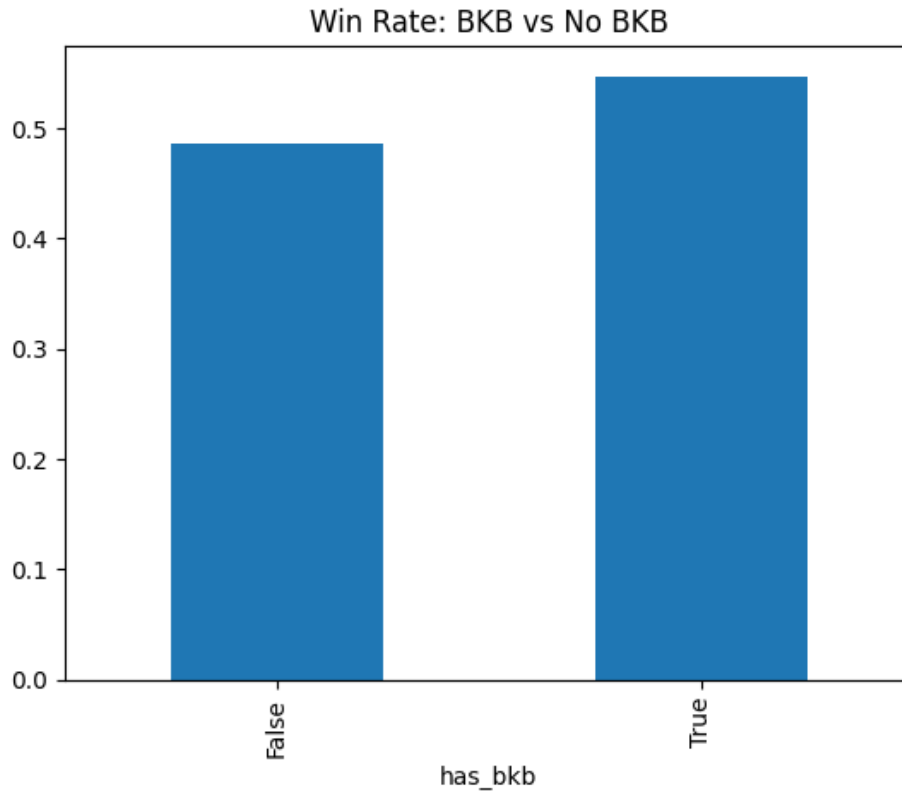


Figure 1: Win rate comparison between players with and without BKB.

The initial result shows that players who purchased BKB had higher average win rate. However, this does not imply that BKB directly caused the win rate increase, because players who can afford BKB may already be in a stronger economic position.

5.2 BKB Impact Under Gold Advantage Context

We further analyze whether BKB matters more in balanced games or one-sided games. Stomp games are classified using final gold advantage and match outcome.

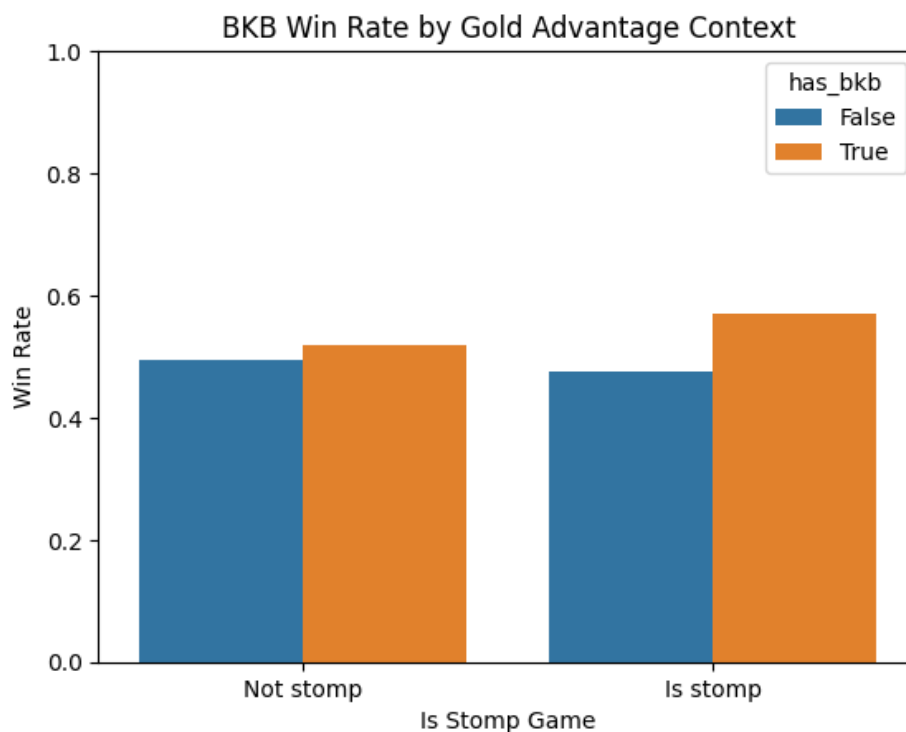


Figure 2: BKB win rate under stomp and non-stomp game contexts.

This analysis highlights that itemization should be interpreted together with game state. Some observed effects may be driven by economic advantage rather than the item alone.

5.3 Carry vs Non-Carry Heroes

We use OpenDota role labels to classify heroes as carry or non-carry. Then we compare BKB win rate across these groups.

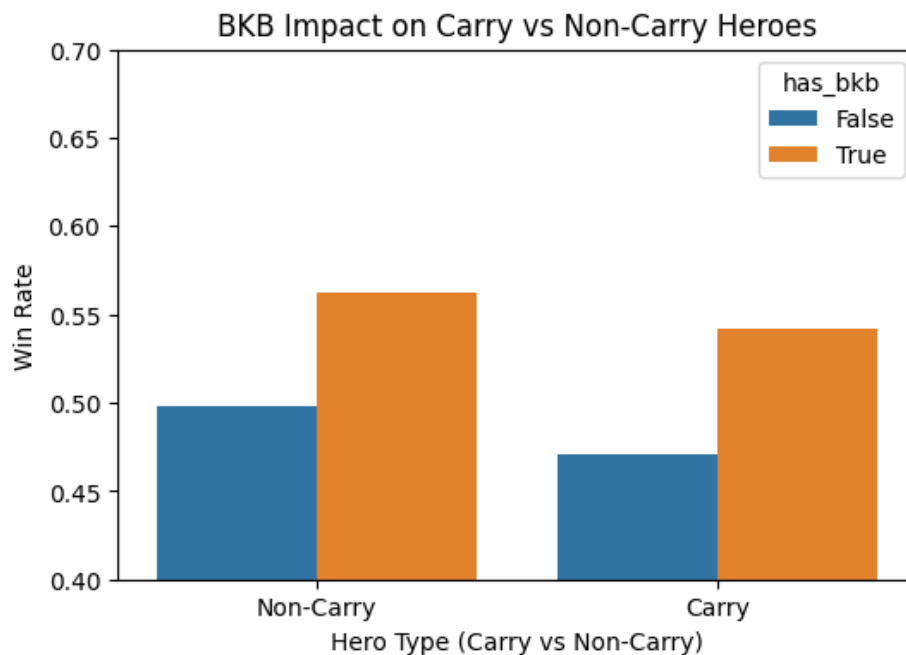


Figure 3: BKB impact on carry and non-carry heroes.

The result shows that both carry and non-carry heroes benefit from BKB. However, support or non-carry heroes may show high BKB impact because they rarely buy BKB unless their team already has a large economic advantage. This is an example of confounding.

5.4 Hero-Level BKB Impact

Next, we calculate BKB impact for each hero.

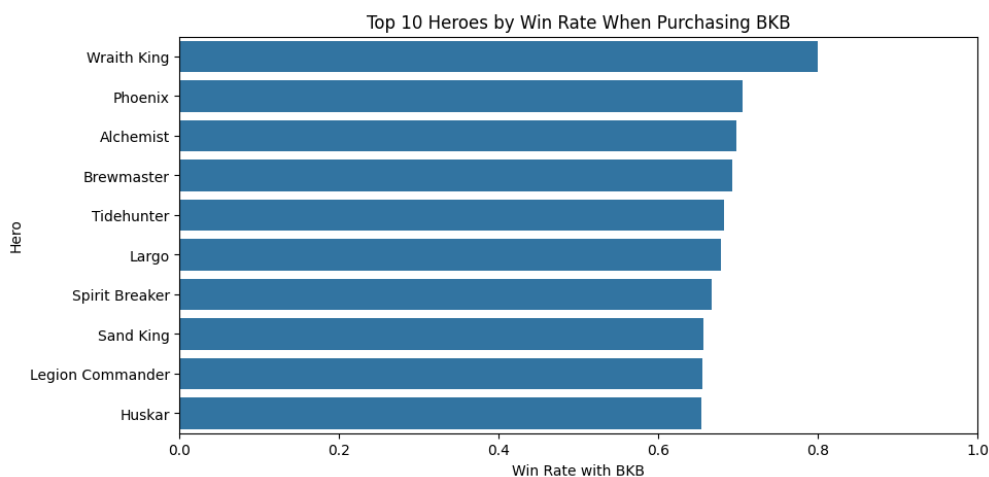


Figure 4: Top heroes when purchasing BKB measured by win rate.

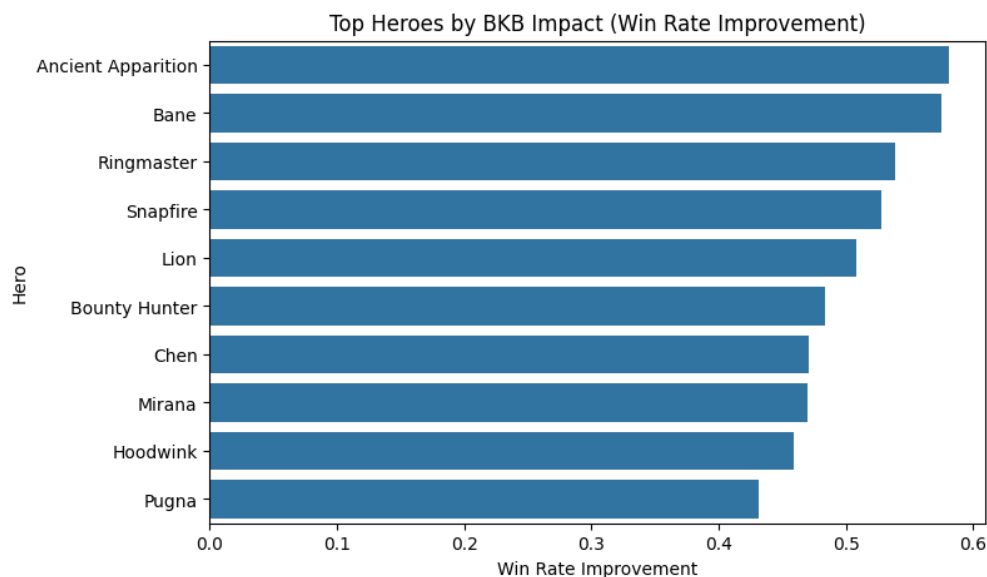


Figure 5: Top heroes by BKB impact measured as win rate improvement.

Several support heroes appear near the top. This does not necessarily mean BKB is best for these heroes. Instead, it likely reflects the fact that when support heroes can afford BKB, their team is often already in a favorable state.

5.5 Matchup-Level BKB Analysis

We then examine BKB impact by hero-enemy pair. Each hero is paired with all enemy heroes in the same match using a self-join.

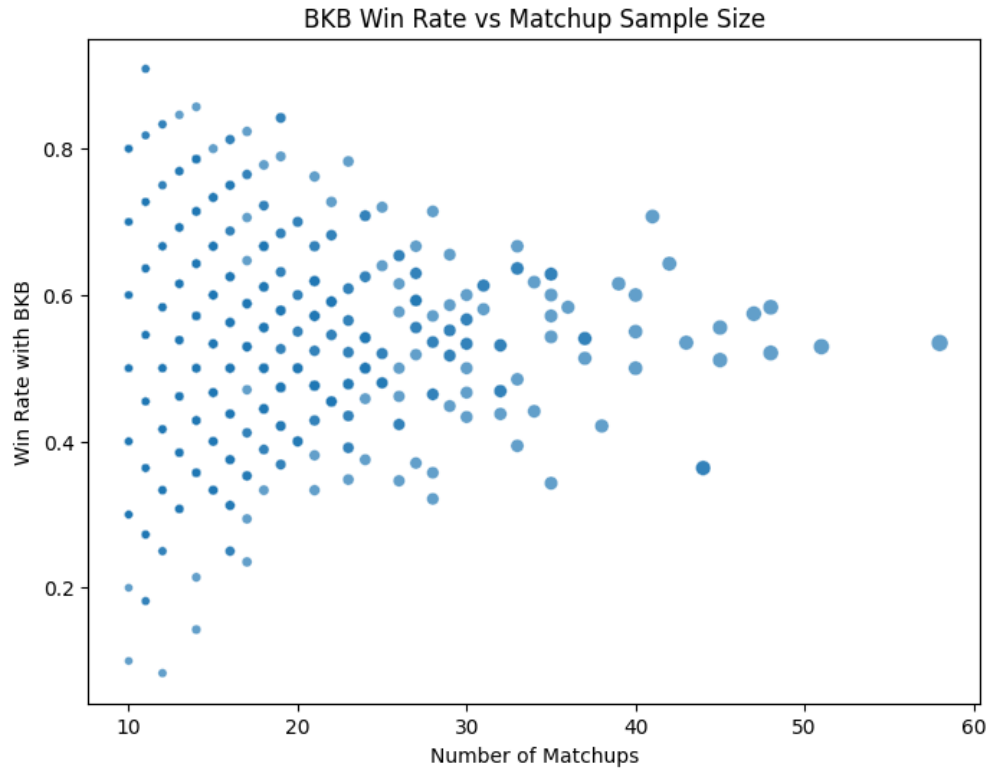


Figure 6: Distribution of BKB win rate across matchup sample-size intervals.

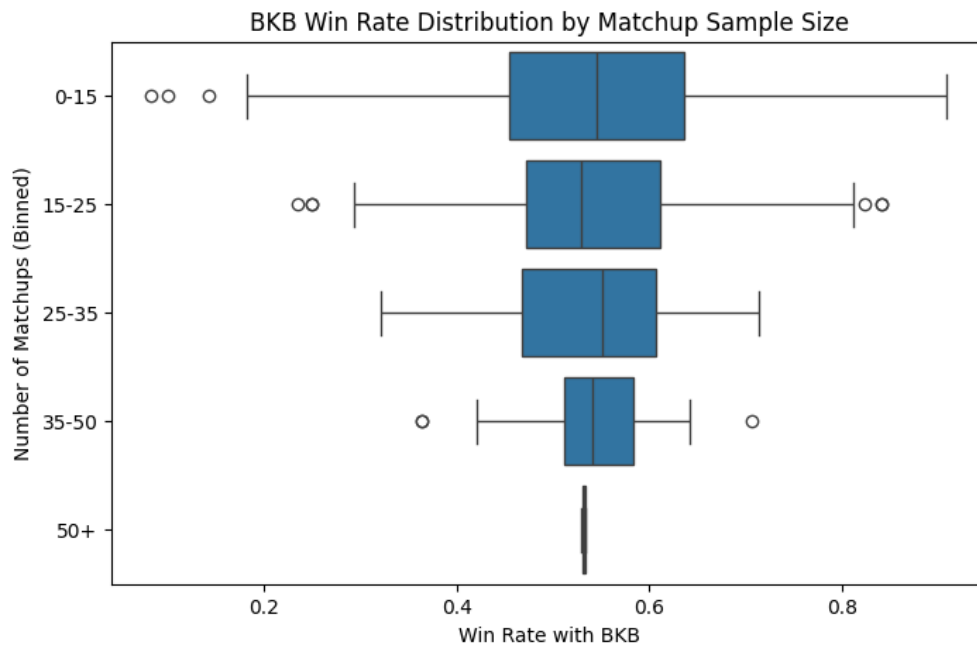


Figure 7: Distribution of BKB win rate across matchup sample-size bins.

The scatterplot and boxplot show that low-sample matchups have wider win-rate distributions,

while larger samples produce more stable estimates. This supports the use of sample thresholds to reduce noise.

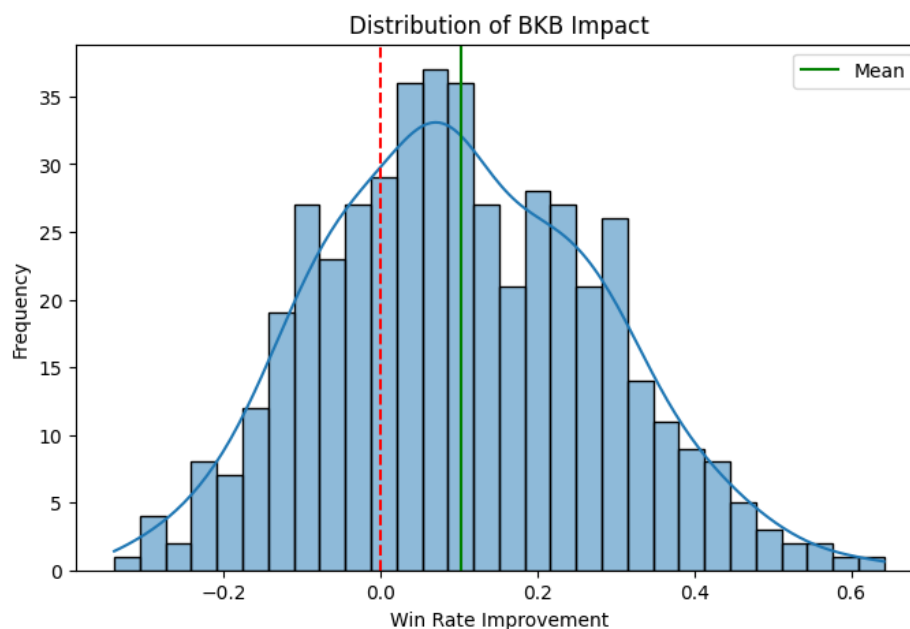


Figure 8: Distribution of BKB impact on win rate across hero–enemy matchups.

The distribution is centered slightly above zero, with a mean improvement of approximately 0.1, indicating that BKB generally provides a moderate positive effect on win rate. However, the presence of negative values suggests that BKB is not universally beneficial and its effectiveness depends on the specific matchup context.

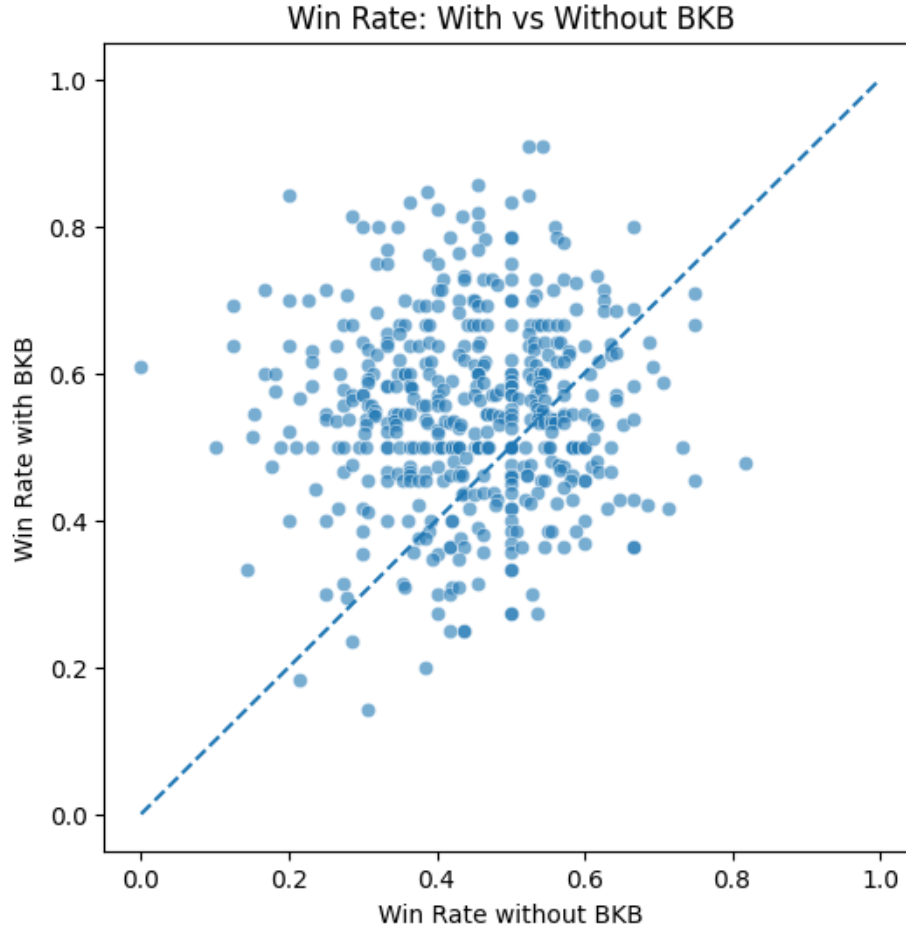


Figure 9: Win rate with BKB versus win rate without BKB. Points above the diagonal indicate positive BKB impact.

Most points are above the diagonal, suggesting that BKB is generally associated with improved win rate. However, the improvement is moderate and varies by matchup.

5.6 Beyond BKB: Item Impact by Matchup

Finally, we extend the method beyond BKB. Given a specific hero and enemy hero, we evaluate which item is associated with the highest win rate improvement.

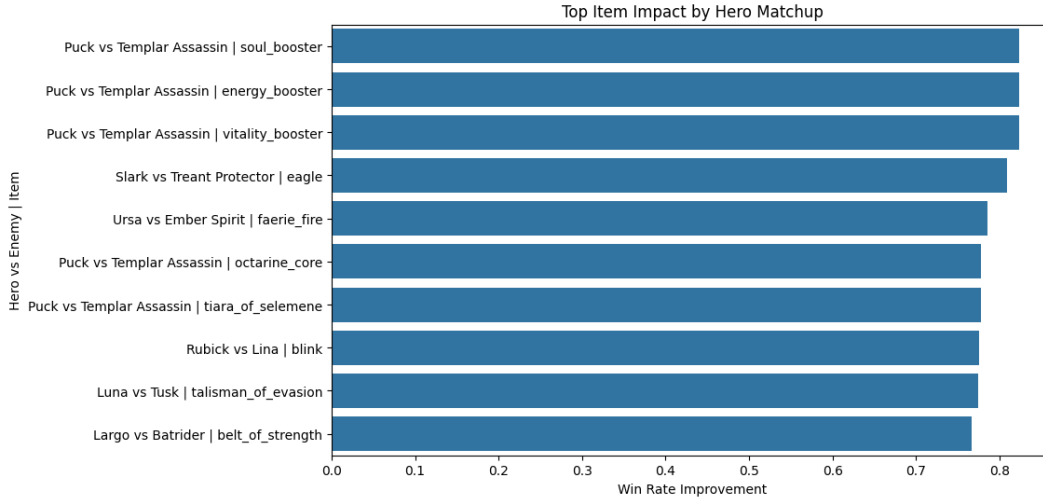


Figure 10: Top item impact by hero-enemy matchup.

This section demonstrates that the framework can be generalized from a single item to matchup-specific item analysis. It can serve as a foundation for future item recommendation systems.

6 Technological Challenges

6.1 Parsing Item Purchase Data

One challenge was that the item purchase column looked like a dictionary but was loaded as a string from CSV. This prevented direct use of Spark map functions. We addressed this by converting the string into a JSON-compatible format and parsing it into a Spark MapType. Then we used `map_entries` and `explode` to create item-level rows.

6.2 Joining Data from Multiple Tables

Important features were distributed across different files. For example, player outcomes were in `players.csv`, while gold advantage over time was in `radiant_goldadv.csv`. We had to aggregate gold advantage to match level and merge it back with player-level records in order to apply functions on the dataframe.

6.3 Self-Join Expansion

Constructing hero-enemy matchups required a self-join on `match_id`. This increased the number of records because each player was paired with five enemy heroes. We used PySpark to make this transformation scalable.

6.4 Correlation versus Causation

A major analytical challenge was avoiding overclaiming. Higher win rate with an item does not prove that the item caused the win. We addressed this by discussing confounding factors such as economic advantage, hero strength, and matchup advantage.

7 Changes in Technology

In the original proposal, we considered using Hadoop, Hive, MongoDB, and machine learning models. During implementation, we focused on Python, Pandas, PySpark, and visualization tools.

This change was made because the available dataset was structured CSV data and did not require a full NoSQL database or Hive warehouse. PySpark was more directly useful for scalable joins, grouping, and aggregation. Machine learning was moved to future work because the main challenge was first to build reliable features and interpret item impact correctly.

8 Uncovered Aspects from Presentation

Due to the 15-minute presentation limit, we could not cover every experiment in detail. In particular, the following aspects were only briefly discussed:

- Why raw item win rate can be misleading.
- How support heroes create selection bias in BKB impact analysis.
- How sample size affects matchup-level win-rate reliability.
- How the same framework can be extended from BKB to all purchased items.

Screenshots of these additional results are included in the Results and Analysis section.

9 Lessons Learned

This project taught us that data analysis in esports requires both technical processing and domain interpretation. A simple win-rate comparison can be misleading if we ignore confounding factors such as gold advantage, hero role, and matchup context.

We also learned the importance of scalable data transformation. Operations such as self-joins and item explosion can quickly increase data size, making Spark-style processing useful even when the original dataset is moderate.

Finally, we learned that careful feature definition is critical. The definition of BKB impact, stomp games, and matchup records directly affects the interpretation of results.

10 Future Improvements

If we had more time, we would extend this project in several directions:

- Compare itemization strategy trends across different months and game patches.
- Given a specific hero and enemy hero composition, identify which item leads to the highest win rate.
- Given a specific hero and enemy hero composition, identify which item combinations lead to the highest win rate.
- Given a specific hero and enemy hero composition, estimate the optimal timing to purchase a specific item.
- Control for confounding factors such as gold advantage, game duration, and player role.
- Build a supervised learning model to predict win probability from hero, enemy lineup, and item choices.

11 References

1. Kaggle Dataset: Dota 2 Pro League Matches 2016–2026. <https://www.kaggle.com/datasets/bwandowando/dota-2-pro-league-matches-2023>
2. OpenDota API Documentation. <https://docs.opendota.com/>
3. OpenDota Hero Metadata API. <https://api.opendota.com/api/heroes>
4. Demediuk, Simon, et al. “Role identification for accurate analysis in Dota 2.” *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, 2019.
5. Semenov, Aleksandr, et al. “Performance of machine learning algorithms in predicting game outcome from drafts in Dota 2.” *International Conference on Analysis of Images, Social Networks and Texts*, 2016.